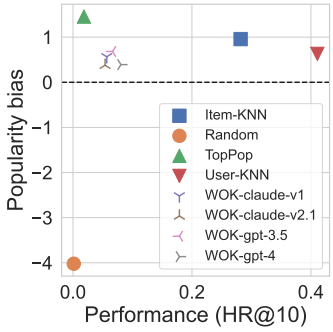


	HR5 ↑	HR10 ↑	popularity bias ↓	unmatched ↓
Random	0.001 ± 0.001	0.001 ± 0.001	−4.020 ± 0.013	0
TopPop	0.005 ± 0.001	0.014 ± 0.002	1.455 ± 0.010	0
ItemKNN	0.204 ± 0.006	0.282 ± 0.006	0.957 ± 0.009	0
UserKNN	**0.313 ± 0.007**	**0.411 ± 0.007**	0.630 ± 0.010	0
WOK-claude-v1	0.044 ± 0.003	0.057 ± 0.003	0.565 ± 0.010	1.360 ± 0.016
WOK-claude-v2.1	0.042 ± 0.003	0.054 ± 0.003	**0.377 ± 0.010**	1.361 ± 0.018
WOK-gpt-3.5	0.047 ± 0.003	0.067 ± 0.004	0.682 ± 0.010	0.613 ± 0.013
WOK-gpt-4	0.050 ± 0.003	0.081 ± 0.004	0.392 ± 0.008	0.869 ± 0.013

Figure 2 & Table 2: Results on a subsample of the MovieLens 10M dataset. We repeat the experiments 5 times, with 1000 users in each fold. Reported are the mean plus/minus one standard error of the mean.

this additional information available to the WOK models by fine-tuning them or infusing the user-item interactions via Retrieval-Augmented Generation (RAG, [36]).

4.5 Popularity bias mitigation and minimization via prompting

A range of mitigation strategies have recently been explored for different types of biases in LLMs. Interventions target the training data [61, 63], the learned representations [10, 45] or employ fine-tuning [48]. However, a surprisingly simple approach has been shown effective [40]. The model can be asked to 'self-debias' [47] via its prompt. In our case, we attempt to leverage the model's world knowledge regarding whether or not a movie is popular, and then ask it to focus on less popular items for debiasing. If successful, this mitigation strategy would be greatly beneficial from a usability perspective: Unlike other methods, it allows a user to configure the desired level of popularity via a natural language interface – no ML expertise required.

Mitigation. To evaluate this method, we add the additional instruction "*Recommend movies that match the average popularity level of the movies the user watched in the past. For instance, if the user mostly watched blockbusters, you should recommend movies that are also blockbusters. If, on the other hand, the user watched less well-known movies, you should recommend niche movies.* " to each of the WOK recommenders. Every such recommender is marked by a *-mitigate* suffix.

We use the same experimental setup as described in Section 4.3. Table 3 and Figure 3 show the results. The popularity bias of all *mitigate*-models (hollow triangles in Figure 3) is slightly decreased when compared to the respective base models ("tripods" in Figure 3). For all base-LLMs apart from claude-v1, this reduction in bias is traded off against a reduction in prediction performance.

Note with the exception of claude-v2.1, the popularity bias values remain relatively close to their base values, or in other words, the mitigation strategy is not highly effective. We therefore also try a second, more extreme mitigation strategy to study how far we can push the WOK recommender into providing long tail recommendations.

Minimization. To this aim, we replace the *mitigate*-instruction by the following instruction: "*Recommend indie, niche, or less well-known movies, avoiding mainstream blockbusters.*" The resulting models are marked by a *-minimize* suffix.

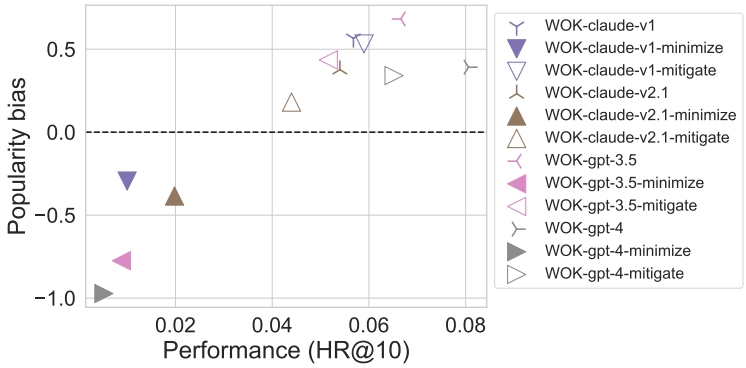
The results are again shown in Table 3 and Figure 3. The *-minimize* strategy does indeed reduce popularity bias to a point where the resulting models show strong negative popularity bias. However, this strategy also results in a strongly decreased recommendation accuracy across all LLMs.

Note also that the number of unmatched items increases considerably when using the mitigation prompt (except for claude-v2.1), which highlights an important limitation of this experiment. Recall that the MovieLens data sets contains "only" 10,000 movies and that WOK recommenders do not have access to this list of valid movies. As a consequence, WOK models that aim to minimize popularity bias might be punished for recommending niche movies that do not appear in the MovieLens catalogue, but could nevertheless be relevant to the user.

4.6 Correlation of popularity bias metrics

We compare the metrics from Table 1 with each other based on the popularity bias values that are reported in Table 2 (8 values, original model performance for each metric) and Table 3 (8 additional values, model performance under two mitigation strategies for each metric). In total we base the correlation analysis on 8+8 = 16 popularity bias values for four different metrics, i.e., Gini−diversity, Herfindahl−diversity, average popularity lift, and our suggested metric, the log popularity difference. Unsurprisingly, log popularity difference and average popularity lift show a very strong correlation, as measured by Kendal's τ [33], as shown in Figure 4. These two metrics are strongly correlated, as both metrics are very similar in nature, but average popularity lift does not satisfy the Desiderata (1), (3) and (4). For an analysis of the practical implications of this, see Appendix B.

Herfindahl− and Gini−diversity measure fundamentally different things than popularity lift and log popularity difference (the difference of popularity *diversity* between user and recommendation vs. the difference of *average popularity* between user and recommendation). Hence, the two families of metrics are not expected



	HR@5 ↑	HR@10 ↑	popularity bias ↓	unmatched ↓
WOK-claude-v1	0.044 ± 0.003	0.057 ± 0.003	0.565 ± 0.010	1.360 ± 0.016
WOK-claude-v1-minimize	0.008 ± 0.001	0.010 ± 0.001	−0.295 ± 0.009	2.001 ± 0.031
WOK-claude-v1-mitigate	0.046 ± 0.003	0.059 ± 0.003	0.532 ± 0.010	1.118 ± 0.014
WOK-claude-v2.1	0.042 ± 0.003	0.054 ± 0.003	0.377 ± 0.000	1.361 ± 0.018
WOK-claude-v2.1-minimize	0.015 ± 0.002	0.020 ± 0.002	−0.385 ± 0.010	0.787 ± 0.016
WOK-claude-v2.1-mitigate	0.033 ± 0.003	0.044 ± 0.003	**0.181 ±0.010**	0.792 ± 0.012
WOK-gpt-3.5	0.047 ± 0.003	0.067 ± 0.004	0.682 ± 0.010	**0.613 ± 0.013**
WOK-gpt-3.5-minimize	0.007 ± 0.001	0.009 ± 0.001	−0.775 ± 0.013	3.393 ± 0.031
WOK-gpt-3.5-mitigate	0.037 ± 0.003	0.052 ± 0.003	0.438 ± 0.012	1.105 ± 0.022
WOK-gpt-4	**0.050 ± 0.003**	**0.081 ± 0.004**	0.392 ± 0.008	0.869 ± 0.013
WOK-gpt-4-minimize	0.004 ± 0.001	0.005 ± 0.001	−0.972 ± 0.009	1.549 ± 0.019
WOK-gpt-4-mitigate	0.045 ± 0.003	0.0652 ± 0.003	0.341 ± 0.007	0.914 ± 0.012

Figure 3 & Table 3: Results for the mitigation experiment. The results are grouped by base LLM in the WOK model. Bold numbers indicate best performance across all models.

to correlate well. In practice, we observe them to be negatively correlated in Figure 4.

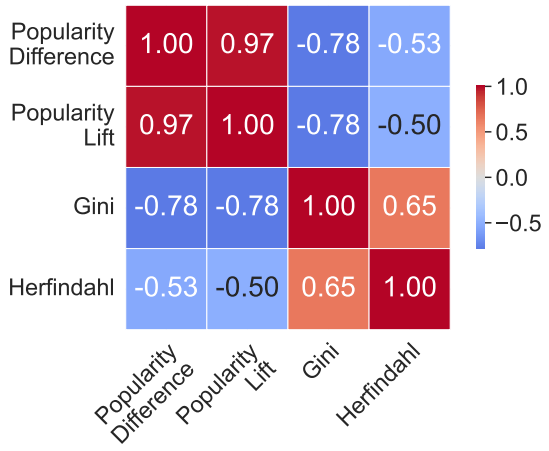


Figure 4: Kendall's Tau correlation coefficients between the various metrics measured across experiments reported in Tables 2 and 3.

5 DISCUSSION AND FURTHER WORK

Recall that our goal was to study popularity bias in recommenders using general-purpose LLMs to evaluate their suitability as off-the-shelf recommenders, as opposed to constructing a state-of-the-art recommender in terms of recommendation accuracy. We therefore deliberately kept the LLM recommender as light-weight as possible to ensure that our findings are reflective of the intrinsic bias in LLMs, rather than being confounded by complex, model-specific factors. In doing so, we aim to provide insights that are broadly applicable across various LLM implementations, contributing to a more generalized understanding of popularity bias in these systems. Nevertheless, the recommendation accuracy of the WOK recommender could likely be improved using various techniques, including 1) fine-tuning on a training set of successful recommendations; 2) "grounding" the recommendations in a movie catalog to avoid hallucinations (via RAG, [36]); or 3) more advanced prompting techniques such as chain-of-thought prompting, or more advanced prompt optimization techniques [19, 55]. Furthermore, these or similar automatic prompt optimization techniques could also be applied to find more balanced prompt-based mitigation strategies that find a happy medium between the two strategies we experimented with.

An open research question is the evaluation of intent-based performance of LLM-based recommenders. As discussed in Section 2.2, easily generating recommendations that are guided by user-intent is a compelling promise of LLM-based RS. In order to evaluate the

performance in this task, one would require structured ground truth, or, as a starting point, use simple proxy tasks where the intent encodes known metadata such as director, genre, year or similar. Similarly, while research in popularity bias is motivated by "rich-get-richer" effects over time, we have not modeled or measured temporal dynamics (see [32, 38, 57] for dynamical modeling approaches, and [49] for an overview).

6 CONCLUSION

We have investigated the promise of LLM-based recommender systems focusing on a specific aspect of their performance and usability: popularity bias. We started by constructing a measurement for the phenomenon. To do so, we have formulated desiderata for a popularity bias metric, and have evaluated existing metrics against these desiderata. Applying some adjustments to the Popularity Lift metric by Abdollahpouri et al. [2], we have arrived at our metric: log popularity difference. We acknowledge that this metric may not suit all future studies on popularity bias given the topic's diverse application domains and goals. However, we encourage future research to examine the assumptions and theoretical properties of their chosen metrics, with our framework potentially serving as a useful starting point.

Using our metric, we have compared traditional RS against the LLM-based models on the MovieLens 10M dataset. We have found the LLM-based recommenders to have moderate amounts of popularity bias; usually less than their traditional, collaborative filtering-based counterparts. In our mitigation experiment, we have found that it is possible to lower popularity bias further by including additional instructions in the prompt. In the extreme case, where the model is specifically instructed to 'avoid mainstream blockbusters', we achieve negative popularity bias. This is accompanied by a drop in recommendation accuracy, which overall is lower for our naïve LLM-based recommenders compared to collaborative filtering baselines.

We believe that LLM-based recommender systems will see widespread usage. We encourage practitioners to measure the popularity bias of such models, and, especially in light of its simplicity, experiment with popularity-debiasing via prompting before deploying such a system.

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